

**SEMINAR SUMMARY:
MARCH 17, 2026 - CHENG LY
EARNING A PH.D.**

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We began the lecture talking about Neuroscience and talking about how the nervous system is electrical in nature. Clearly our nervous system has something electrical going on with it, because for example defibrillators rely on the application of a voltage potential to help reset the heart. We then looked at another example where only electrical stimuli are harnessed in a way as to control a robotic apparatus (an arm for example). So not only are electrical potentials used for bodily regulation, they are in fact used in all facets of both the somatic and autonomic nervous system (willful control versus involuntary functions).

After the introduction of the electrical nature of the nervous system, we went into discussing various different collaborators of the student variety that Dr. Ly had. The list was extensive and across various institutions which was the most impressive bit of it all. Furthermore, there was a good representation of all different types of students from undergraduates to Ph.D students.

We then explored mechanics of working with Dr. Ly. He predominantly works using coupled networks as his mathematical modeling technique while using Matlab as the computational tool behind the coupled networks. There were a few other tools mentioned: C, XPP, and python, for computational purposes.

Interestingly on the subsequent slide to talking about the various different techniques used for doing his research, Dr. Ly presented the anatomy of a nerve cell with the soma (body of the cell), the axon (the wire of the cell), and the dendrites (the chemoreceptors that kick off a transference of electric potential down the axon). He then put a graph of the electric potential over time in voltage displaying a steady state with large spikes in voltage (which would correlate to the chemical stimuli, sodium active transport).

The next slide was even more interesting. The slide displayed a graph of nerve cell voltage spikes as related to the position of a black bar in the visual field. Further, he had a map of the visual cortex (taken from the occipital lobe of the brain) such that we could see which neurons were firing under which rotations, and the results varied wildly indicating that there is a division of labour that the nerve cells adhere to, though it is too complex to fully mathematically model properly. Nonetheless, it is there, and that's pretty cool. Lastly, in this area we noted that the response we get is not consistent from experiment to experiment.

So, the division of labour is there, but it fundamentally changes over time, which is even wilder.

We then went to explore neural oscillations based upon the Hodgkin-Huxley model for describing action potentials in neurons. We had a graph of spiking periodic voltage versus time, and were given that the model depends on:

$$\begin{aligned}\frac{d\Theta}{dt} &= \omega, & \Theta(t^-) = 1 &\Rightarrow \Theta(t^+) = 0 \\ \frac{d\Theta}{dt} &= \omega + \Delta(\Omega)[\dots\text{perturbations}\dots].\end{aligned}$$

We then talked for a bit about Phase Resetting Curves. We went over a good deal more here about different physiological electrical mechanisms in the body worth modeling and some models of them.

Lastly we went over an article on how to pick a graduate advisor. The article, if we were to summarize it, said that one need consider two criteria when picking a graduate advisor: scientific ability and mentorship ability. Furthermore, it is important that the advisor to specialize in the same area as the student has interest in. Listening to other faculty about whom to pick is important; leaning on resources that you are already familiar with. Research should be fun such that you actually want to spend hours pining over the puzzle of a problem. It's all quite exciting.